Huffman Coding

Solution:— "DATAANALYTICSANDINTELLIGENCELABORATORY"

1.] Character and frequency of the Given text message

B = G = S = 1

C = D = O = R = Y = 2 [Total Character = 38]

E = I = 3

L = N = T = 4

A = 7

2.] Arrange in the increasing order of the frequency.

- B=1, G=1, S=1 ✓
- C=2, D=2, O=2, R=2, Y=2 ✓
- E=3, I=3 ✓
- L=4, N=4, T=4 ✓
- A=7

Important:—

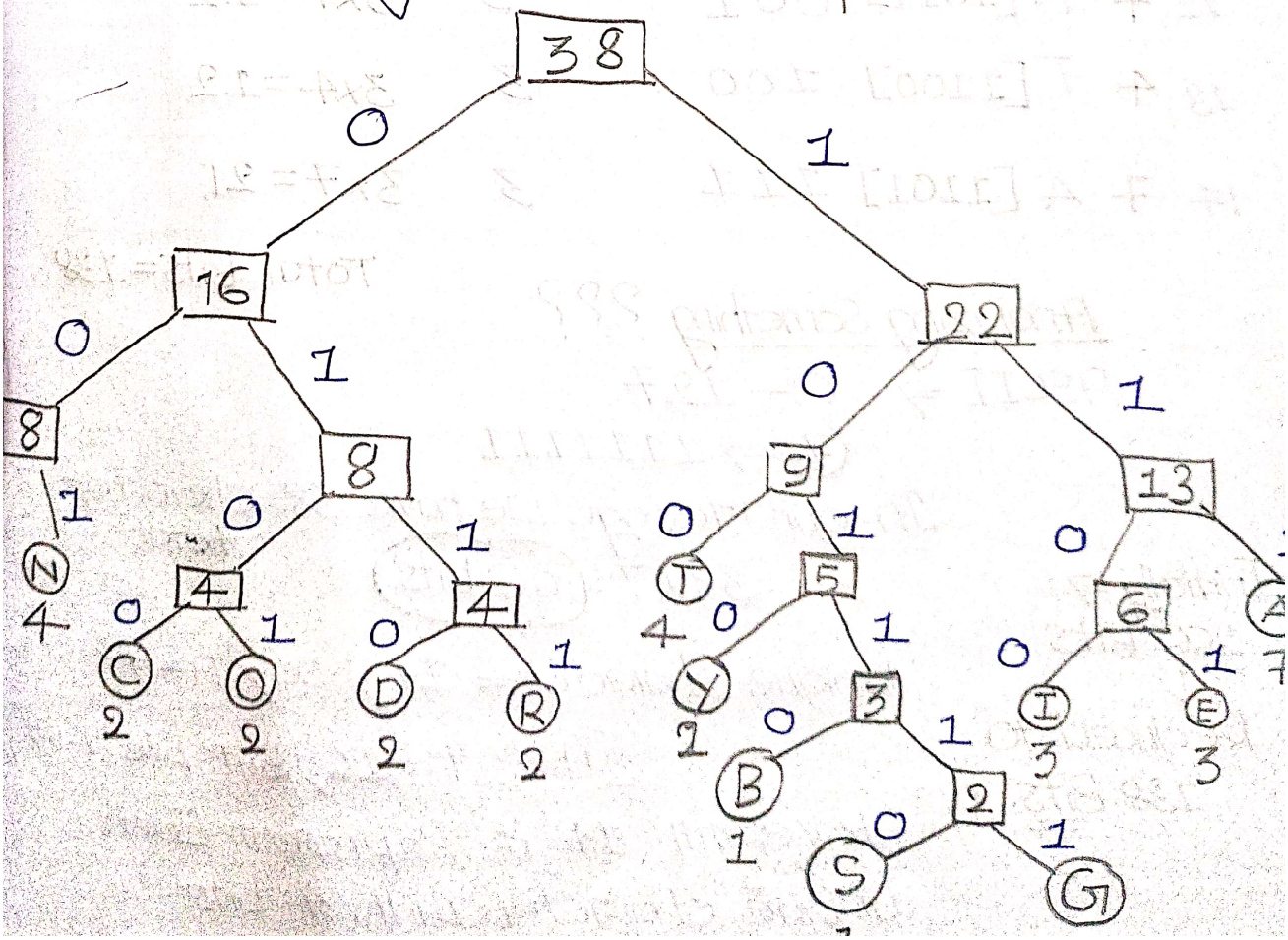
TT = transmission time

$TT = \frac{MS}{BW}$ → Message size, ↳ Band width

[TT ∝ MS]

3.] Construct a Huffman Tree

Taking ⇒ L = 0, R = 1



	Character (f)	[4 Bits]	Code	Code length	[Bits calculation]
1	1	B[0000]	10110	5	5×1=5
2	1	G[0001]	101111	6	6×1=6
3	1	S[0010]	101110	6	6×1=6
4	2	C[0011]	0100	4	4×2=8
5	2	D[0100]	0110	4	4×2=8
6	2	O[0101]	0101	4	4×2=8
7	2	R[0110]	0111	4	4×2=8
8	2	Y[0111]	1010	4	4×2=8
9	3	E[1000]	1101	4	4×3=12
10	3	I[1001]	1100	4	4×3=12
11	4	L[1010]	000	3	3×4=12
12	4	N[1011]	001	3	3×4=12
13	4	T[1100]	100	3	3×4=12
14	7	A[1101]	111	3	3×7=21

Total Bits = 138

Analyzing Something ???

ASCII ⇒ 0 – 127

7 → 1111111

In our Message we have 38 character total.

38×7 = 266 bits.

Initially ⇒ 266 Bits → Reduced to 138 Bits.

Unique character = 14, say n = 4

$2^n = 2^4 = 16$, But we will take till 14 & Represent Each unique character with 4 bits.

Average Bits required to represent each character = $\frac{138}{38} = 3.6$